

חלק א' - ניתוח דרישות ועיצוב ראשוני של המערכת
מודל מלאי

1. Requirements

1.1 Table 1 – Requirements

ID	Functional/N on-Functional	Description	Priority	Risk	Status
1.	Functional	The system must store for each product: name, unique ID, and manufacturer.	MH	Low	Done
2.	Functional	The system must store the location of each inventory item.	MH	Low	Done
3.	Functional	The system must store for each item its cost price and selling price.	MH	Low	Done
4.	Functional	The system shall allow the assignment of every inventory item to a specific category.	MH	Low	Done
4.	Functional	The system must maintain two quantity fields for each item: shelf quantity and warehouse quantity.	MH	Low	Done
5.	Functional	The system must allow adding a new item to the inventory with all its details	MH	Low	Done
6.	Functional	The system must allow updating the quantity of an existing item on the shelf or in the warehouse	MH	Low	Done
7.	Functional	The system must allow transferring a quantity of an item from the warehouse to the store shelves	MH	Low	Done
8.	Functional	The system must allow setting a minimum quantity threshold for each item	MH	Low	Done
9.	Functional	The system must alert when the total quantity of an item reaches or falls below its	MH	High	Done

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		minimum threshold			
	Functional	The system must allow generating an order report showing all items that need to be restocked based on current inventory levels	MH	Low	Done
10.	Functional	The system must store the supplier discount for each item as part of the item's record	MH	Low	Done
11.	Functional	The system must allow defining a promotion containing a discount percentage on a specific item or category, with a start date and end date	MH	Low	Done
12.	Functional	The system must organize items in a hierarchical category tree	MH	High	Done
13.	Functional	The system must allow generating an inventory report filtered by one or more categories, on demand	MH	Low	Done
14.	Functional	The system must allow an employee to mark an item as damaged or expired	MH	Low	Done
15.	Functional	The system must track all items marked as damaged or expired	MH	High	Done
16.	Functional	The system must allow generating a periodic report of all damaged and expired items	MH	Low	Done
17.	Functional	The system must allow searching for an item by name or category	MH	Low	Done
18.	Functional	The system must allow removing an item from the inventory	MH	Low	Done
19.	Functional	The system must dynamically calculate the final selling price of an item	MH	High	Done
20.	Functional	The system must allow moving	MH	Low	Done

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		items from store shelves back to the warehouse			
21.	Functional	The system must allow removing all defective (damaged or expired) items from inventory at once	MH	Low	Done
22.	Non - Functional	The system shall use a layered architecture (UI, Service, Domain) with SRP, keeping business logic fully decoupled from the UI so the CLI can be replaced with a GUI without changing Domain layer code.	MH	High	Done
22.	Non - Functional	The system shall never crash on invalid user input. All errors shall be handled via Java Exceptions/Try-Catch, returning the user to the main menu.	MH	Low	Done

2.2 Open Questions for Table 2

#	Topic	Question	answer
1.	Stock alert	Is the alert triggered based on total quantity (shelf + warehouse) or shelf quantity only?	The alert is triggered based on the total quantity (shelf + warehouse combined) based on the forum
2.	Promotions	If two promotions apply to the same item simultaneously, which one takes priority?	only the most beneficial promotion for the customer will be applied based on the forum
3.	Item Location Detail	Does the item location refer to its position in the store, in the warehouse, or both? And what level of detail is required for the location	If the item is in the warehouse, the location is simply "warehouse". If the item is in the store, the location must specify the aisle and shelf number. It can be assumed that each branch has only one warehouse. based on the forum
4.	Inventory Alert Delivery	How should the system deliver low stock alerts, and to whom should they be delivered?	The alert will be displayed as a printed message on screen to whoever is currently operating the system

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5.	Minimum Threshold Calculation	How exactly is the minimum quantity threshold calculated?	The minimum quantity threshold is set manually. The system will allow the user (manager) to enter this value manually when creating a new item or update it afterward.
6.	Damaged/Expired Item Tracking Details	What exact information needs to be stored for damaged/expired items?	A damaged/expired record will store: item name, quantity affected, date reported, status (damaged or expired), and location (warehouse/aisle+shelf)"
7.	Tracking Cost Prices & Supplier Discounts	To track supplier discounts, must the system maintain a historical price log, or is storing only the current cost and selling prices sufficient?"	storing only the current cost and selling prices
8.	Item Expiration Date	Are expiration dates tracked globally per item, or individually by unit/batch?	per physical unit
9.	The Shortage & Purchasing Report	What exact data should be displayed in the report?	must include the following fields for each item: product name, SKU (catalog number), quantity, and location (warehouse/shelves). ● based on the forum
10.	The Category Inventory Report	What data columns should be displayed in the report?	The report will display: Item Name, Sub-Category, Shelf Quantity, Warehouse Quantity, Total Quantity, Cost Price, and Selling Price.
11.	Category Hierarchy Depth Limit	must it always be exactly 3 levels?	the system must support a flexible, dynamic hierarchy (N-levels) and should not be hardcoded to exactly 3 levels. ● based on the forum

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